CMDA-4654
Project 1

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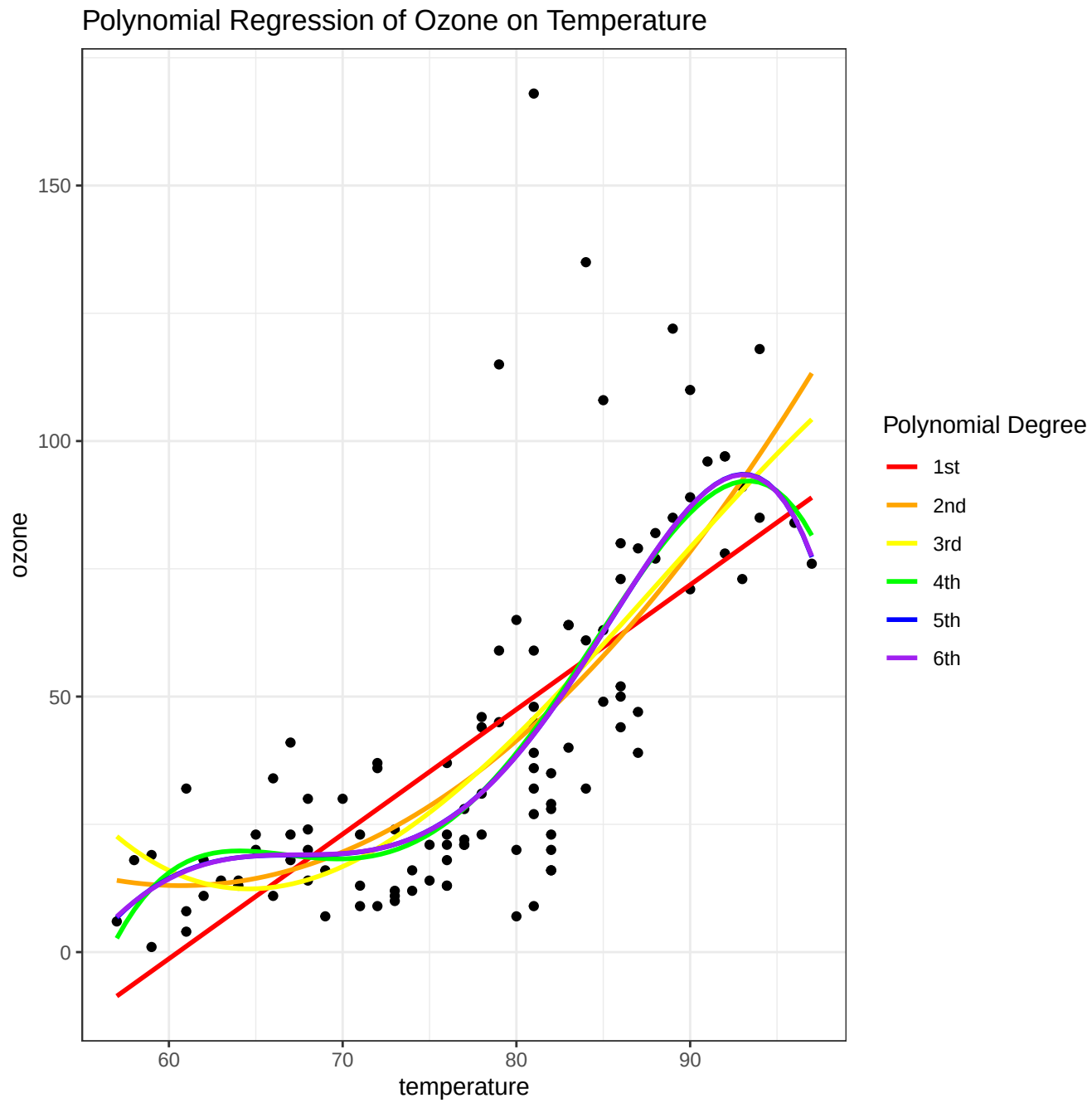
```
library(ggplot2)
library(tidyverse)
library(kableExtra)
library(patchwork)
source("Reece_Wong_Zahir_Project_1.R")
```

Problem 1

```r
# Loading in the data set
load("ozone.RData")
data("ozone")
```

Part 1

```r
# Fit models and overlay on top of each other
# For easy comparison
ggplot(ozone, aes(x = temperature, y = ozone)) + theme_bw() + geom_point() +
  geom_smooth(method = "lm", se = F, aes(color = "1st")) +
  geom_smooth(method = "lm", formula = "y ~ poly(x, 2)", se = F, aes(color = "2nd")) +
  geom_smooth(method = "lm", formula = "y ~ poly(x, 3)", se = F, aes(color = "3rd")) +
  geom_smooth(method = "lm", formula = "y ~ poly(x, 4)", se = F, aes(color = "4th")) +
  geom_smooth(method = "lm", formula = "y ~ poly(x, 5)", se = F, aes(color = "5th")) +
  geom_smooth(method = "lm", formula = "y ~ poly(x, 6)", se = F, aes(color = "6th")) +
  scale_color_manual(name = "Polynomial Degree",
                     breaks = c("1st", "2nd", "3rd", "4th", "5th", "6th"),
                     values = c("red", "orange", "yellow", "green", "blue", "purple")) +
  labs(title = "Polynomial Regression of Ozone on Temperature")
```



From the graph, it seems like the 2nd-degree fit is the best. However, none of the degrees seem to give a satisfactory fit.

Part 2

```
# Where we'll store all the LOESS fits we're making
models <- vector(mode = "list", length = 20)
# Staring span value
span_val <- 0.25
# Fit all the degree-1 models
for (i in 1:10) {
  models[[i]] <- myloess(x_train = ozone$temperature, y_train = ozone$ozone,
                         span = span_val, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Reset span
span_val <- 0.25
# Fit all the degree-2 models
for (i in 11:20) {
```

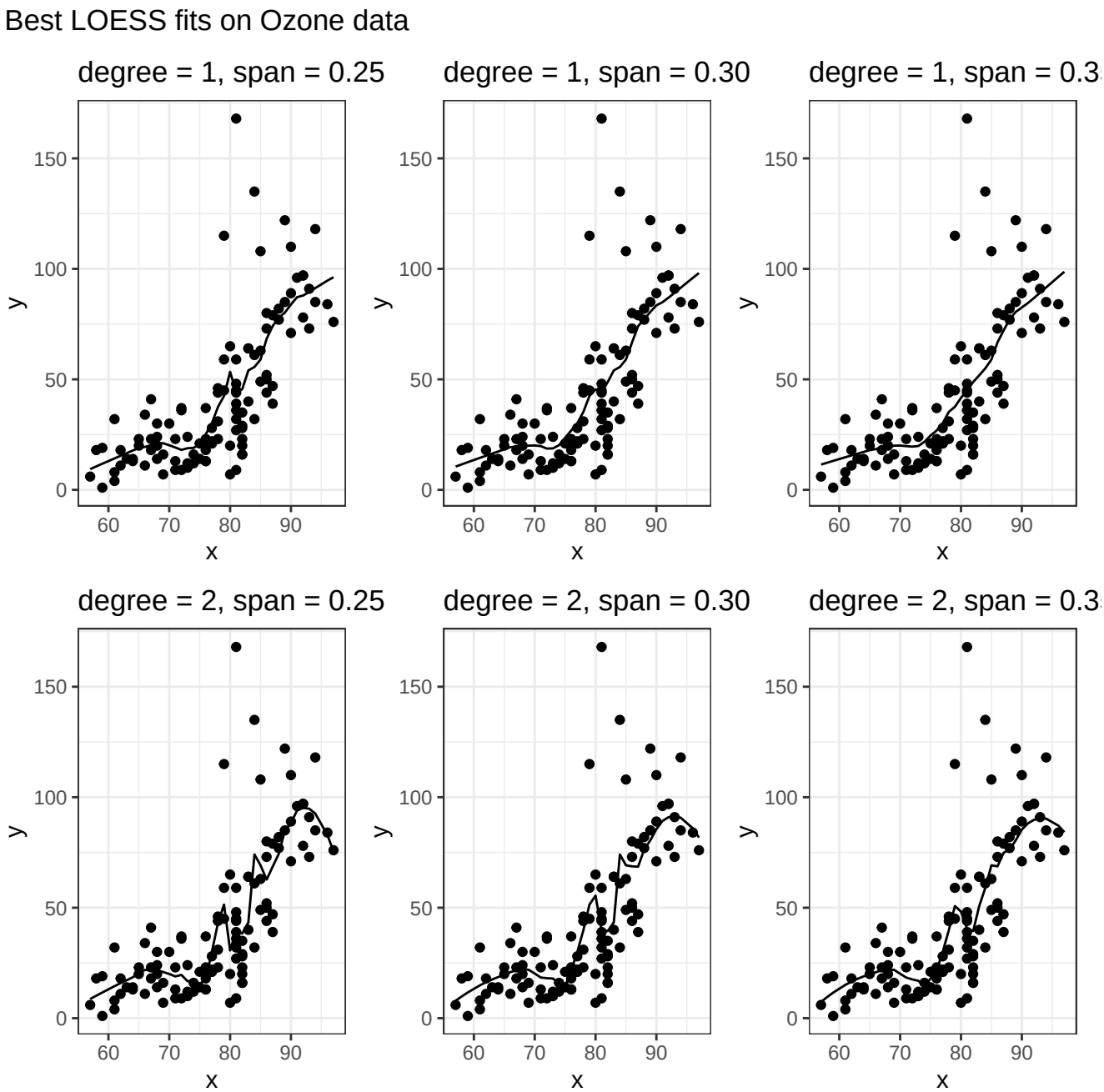
```r
  models[[i]] <- myloess(x_train = ozone$temperature, y_train = ozone$ozone,
                         span = span_val, degree = 2, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Put the span, degree, and MSE into dataframe for easy table making
model_df <- dplyr::bind_rows( lapply(models, function(m) {
  data.frame(
      span = m$span,
      degree = m$degree,
      MSE = m$MSE
    )
  })
)
# Output table
knitr::kable(
  model_df,
  caption = "LOESS Fits on Ozone Data"
)
```

Table 1: LOESS Fits on Ozone Data

span	degree	MSE
0.25	1	467.6651
0.30	1	470.7391
0.35	1	476.7014
0.40	1	485.8182
0.45	1	482.4721
0.50	1	484.9289
0.55	1	486.0865
0.60	1	487.3726
0.65	1	489.6556
0.70	1	491.0533
0.25	2	403.1341
0.30	2	427.5517
0.35	2	428.5621
0.40	2	443.4471
0.45	2	468.4704
0.50	2	471.9893
0.55	2	473.4295
0.60	2	474.5741
0.65	2	475.6940
0.70	2	477.2696

From the table, the best fits with a degree of 1 occur for span values of 0.25, 0.30, and 0.35, and the best fits with a degree of 2 occur for span values of 0.25, 0.30, and 0.35 as well.

```r
# Use the plots from the myloess function, but add titles
p1 <- models[[1]]$loessplot + labs(title = "degree = 1, span = 0.25")
p2 <- models[[2]]$loessplot + labs(title = "degree = 1, span = 0.30")
p3 <- models[[3]]$loessplot + labs(title = "degree = 1, span = 0.35")
p4 <- models[[11]]$loessplot + labs(title = "degree = 2, span = 0.25")
p5 <- models[[12]]$loessplot + labs(title = "degree = 2, span = 0.30")
p6 <- models[[13]]$loessplot + labs(title = "degree = 2, span = 0.35")
# Use patchwork library to plot all together
(p1 + p2 + p3) / (p4 + p5 + p6) + plot_annotation(title = "Best LOESS fits on Ozone data")
```



Based the plots, I would consider all 6 of the models to be better fits than any of the polynomial fits. However, I would also consider almost all of the curves to be somewhat overfitted to the data. In particular, the curves where the local fits used a degree 2 polynomial are very jagged, seemingly in response to some small shifts in the data; the degree-1 models with spans of 0.25 and 0.30 also show some of this jaggedness, though on a lesser degree. Of all of the "best" fits, the one with the least signs of overfitting is the LOESS curve with degree = 1 and span = 0.35.

Part 3

```
# Where we'll store all the LOESS fits we're making
baser_models <- vector(mode = "list", length = 20)
# Staring span value
span_val <- 0.25
# Fit all the degree-1 models
for (i in 1:10) {
  baser_models[[i]] <- loess(ozone ~ temperature, data = ozone, span = span_val, degree = 1)
  span_val <- span_val + 0.05
}
# Reset span
span_val <- 0.25
```

```r
# Fit all the degree-2 models
for (i in 11:20) {
  baser_models[[i]] <- loess(ozone ~ temperature, data = ozone, span = span_val)
  span_val <- span_val + 0.05
}
# Put the span, degree, and RSE into dataframe for easy table making
baser_model_df <- dplyr::bind_rows(lapply(baser_models, function(m) {
  data.frame(
      span = m$pars$span,
      degree = m$pars$degree,
      RSE = m$s
    )
  })
)
# Output table
knitr::kable(
  baser_model_df,
  caption = "Loess Fits on Ozone Data, with loess()"
)
```

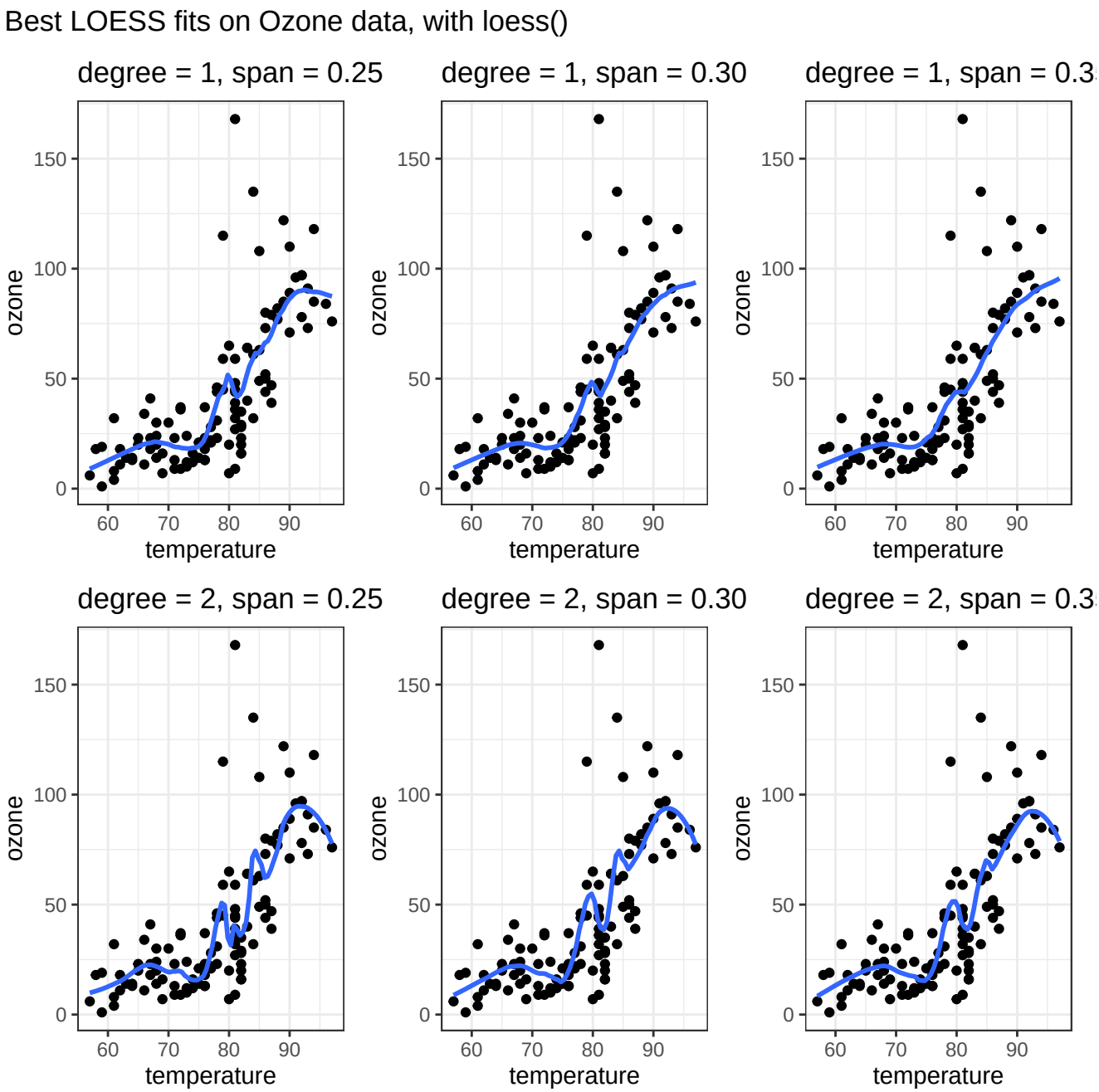
Table 2: Loess Fits on Ozone Data, with loess()

span	degree	RSE
0.25	1	22.03363
0.30	1	22.13759
0.35	1	22.20393
0.40	1	22.35919
0.45	1	22.42632
0.50	1	22.45759
0.55	1	22.42290
0.60	1	22.45481
0.65	1	22.44426
0.70	1	22.44382
0.25	2	21.35833
0.30	2	21.68588
0.35	2	21.67665
0.40	2	21.83337
0.45	2	22.12619
0.50	2	22.19365
0.55	2	22.27206
0.60	2	22.33931
0.65	2	22.36149
0.70	2	22.35995

Although the built in `loess()` function uses RSE instead of MSE, it agrees on the best 3 models of each degree.

```r
# Use the plots from the myloess function, but add titles
g1 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.25, se = F, method.args = list(degree = 1)) +
  labs(title = "degree = 1, span = 0.25")
g2 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.3, se = F, method.args = list(degree = 1)) +
  labs(title = "degree = 1, span = 0.30")
g3 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.35, se = F, method.args = list(degree = 1)) +
  labs(title = "degree = 1, span = 0.35")
g4 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.25, se = F, method.args = list(degree = 2)) +
  labs(title = "degree = 2, span = 0.25")
```

```r
g5 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.3, se = F, method.args = list(degree = 2)) +
  labs(title = "degree = 2, span = 0.30")
g6 <- ggplot(ozone, aes(x = temperature, y = ozone)) + geom_point() + theme_bw() +
  geom_smooth(method = "loess", span = 0.35, se = F, method.args = list(degree = 2)) +
  labs(title = "degree = 2, span = 0.35")
# Use patchwork library to plot all together
(g1 + g2 + g3) / (g4 + g5 + g6) + plot_annotation(title = "Best LOESS fits on Ozone data, with loess()")
```



The plots look very similar to those from the `myloess()` function. The only difference is that the curves look a tad smoother.

Problem 2

```r
# Loading the data
library(MASS)
data("mcycle")
load("mcycle_test.RData")
data("mcycle_test")
```

Part 1

We can use the same fitting-and-plotting process as in part 3 of problem 1:

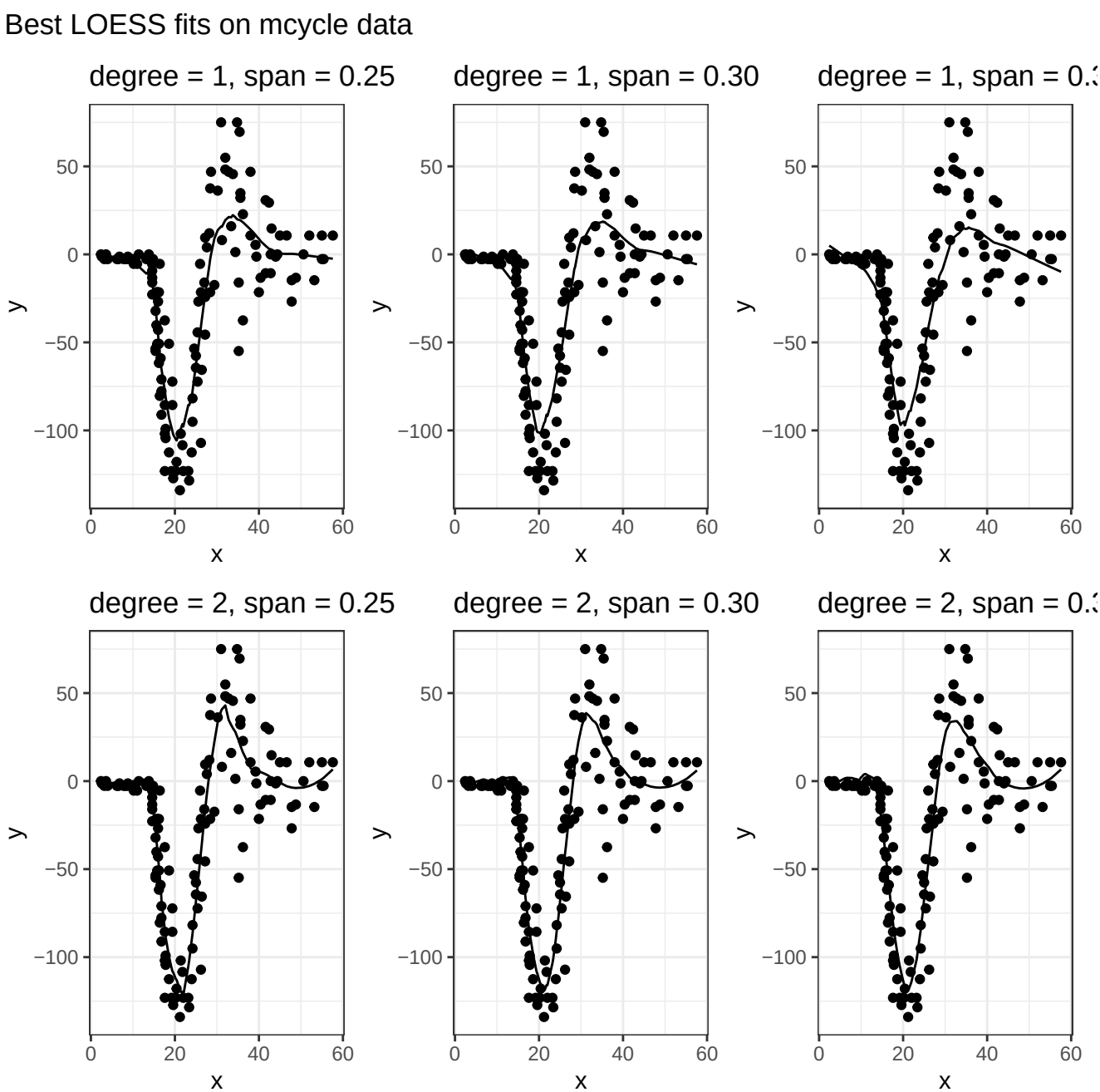
```
# Where we'll store all the LOESS fits we're making
models <- vector(mode = "list", length = 20)
# Staring span value
span_val <- 0.25
# Fit all the degree-1 models
for (i in 1:10) {
  models[[i]] <- myloess(x_train = mcycle$times, y_train = mcycle$accel,
                         span = span_val, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Reset span
span_val <- 0.25
# Fit all the degree-2 models
for (i in 11:20) {
  models[[i]] <- myloess(x_train = mcycle$times, y_train = mcycle$accel,
                         span = span_val, degree = 2, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Put the span, degree, and MSE into dataframe for easy table making
model_df <- dplyr::bind_rows( lapply(models, function(m) {
  data.frame(
      span = m$span,
      degree = m$degree,
      MSE = m$MSE
    )
  })
)
# Output table
knitr::kable(
  model_df,
  caption = "LOESS Fits on mcycle Data"
)
```

Table 3: LOESS Fits on mcycle Data

span	degree	MSE
0.25	1	531.5884
0.30	1	586.2357
0.35	1	664.9955
0.40	1	774.5691
0.45	1	878.0587
0.50	1	1019.2968
0.55	1	1165.2179
0.60	1	1288.7858
0.65	1	1395.3525
0.70	1	1498.9039
0.25	2	455.4035
0.30	2	461.9929
0.35	2	475.7662
0.40	2	504.5805
0.45	2	549.4119
0.50	2	601.6663
0.55	2	674.9693
0.60	2	753.7812
0.65	2	875.4644
0.70	2	1020.6186

For degree = 1, the best span values are 0.25, 0.30, and 0.35. For degree = 2, the best span values are also 0.25, 0.30, and 0.35.

```
# Use the plots from the myloess function, but add titles
p1 <- models[[1]]$loessplot + labs(title = "degree = 1, span = 0.25")
p2 <- models[[2]]$loessplot + labs(title = "degree = 1, span = 0.30")
p3 <- models[[3]]$loessplot + labs(title = "degree = 1, span = 0.35")
p4 <- models[[11]]$loessplot + labs(title = "degree = 2, span = 0.25")
p5 <- models[[12]]$loessplot + labs(title = "degree = 2, span = 0.30")
p6 <- models[[13]]$loessplot + labs(title = "degree = 2, span = 0.35")
# Use patchwork library to plot all together
(p1 + p2 + p3) / (p4 + p5 + p6) + plot_annotation(title = "Best LOESS fits on mcycle data")
```



All of the plots look pretty good, but in my opinion, the models with degree = 2 do a better job capturing the true shape than the models with degree = 1, especially for the flat section of points at the far left and the large curve in the middle of the plot. Also, all of the best models with degree = 2 have lower MSE values than all of the best models with degree = 1. Thus, I would say LOESS with degree = 2 and a span of 0.25, 0.30, or 0.35 provides the best fits to the data.

Part 2

```r
# Where we'll store all the LOESS fits we're making
test_models <- vector(mode = "list", length = 32)
# Staring span value
span_val <- 0.1
# Fit all the degree-1 models
for (i in 1:16) {
  test_models[[i]] <- myloess(x_train = mcycle$times, y_train = mcycle$accel,
                              x_test = mcycle_test$times, y_test = mcycle_test$accel,
                        span = span_val, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Reset span
span_val <- 0.1
# Fit all the degree-2 models
for (i in 17:32) {
  test_models[[i]] <- myloess(x_train = mcycle$times, y_train = mcycle$accel,
                              x_test = mcycle_test$times, y_test = mcycle_test$accel,
                        span = span_val, degree = 2, show.plot = FALSE)
  span_val <- span_val + 0.05
}
# Put the span, degree, and MSE into a dataframe for easy table making
test_model_df <- dplyr::bind_rows( lapply(test_models, function(m) {
  data.frame(
      span = m$span,
      degree = m$degree,
      MSE = m$MSE
    )
  })
)
# Output table
knitr::kable(
  test_model_df,
  caption = "LOESS Fits on mcycle testing data"
)
```

Table 4: LOESS Fits on mcycle testing data

span	degree	MSE
0.10	1	74.45679
0.15	1	77.37762
0.20	1	110.69368
0.25	1	170.22404
0.30	1	241.86712
0.35	1	359.60204
0.40	1	502.30695
0.45	1	645.47300
0.50	1	836.33576
0.55	1	1025.62133
0.60	1	1193.74650
0.65	1	1342.40226
0.70	1	1474.81155
0.75	1	1588.64971
0.80	1	1709.52571
0.85	1	1891.98853
0.10	2	104.61616
0.15	2	79.17058
0.20	2	71.77071
0.25	2	74.31634

span	degree	MSE
0.30	2	80.62825
0.35	2	95.67421
0.40	2	141.93397
0.45	2	193.36285
0.50	2	272.94385
0.55	2	357.88760
0.60	2	433.95078
0.65	2	566.77605
0.70	2	706.61464
0.75	2	848.27038
0.80	2	1007.02030
0.85	2	1174.29534

For both of degree 1 and 2, the optimal spans that minimize the MSE is 0.25.

Problem 3

Problem 4

Part 1

kNN

Part 2

Part 3

Part 4

Part 5

dnkNN

Part 2

Part 3

Part 4

Part 5

Problem 5

Part a

Part b

Problem 6